

The Immersive VR Classroom

| Reflecting on Teamwork, Growth, and
Project Success

Team Members:

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Project Overview

- **Project Summary: Development of an ethical, immersive VR platform for STEM education.**
- **Presentation Focus: A reflection on our team's journey, aligned with the assessment criteria.**

- 1. Team Roles & Responsibilities**
- 2. Skills & Knowledge Development**
- 3. Team Strengths & Weaknesses**
- 4. Team Communication Practices**

- 5. Effective Time Management**
- 6. Fostering Teamwork & Collaboration**
- 7. Internal & External Resource Management**
- 8. Project Outcomes & Reflective Summary**

Team Roles & Responsibilities

➡ **Project Manager:**

Role: Oversaw project timeline, budget, and cross-departmental coordination.

Responsibility: Acted as the primary liaison with university faculty (our "senior management").

➡ **Lead Developer:**

Role: Architected the technical solution and managed the codebase.

Responsibility: Ensured the platform was robust, scalable, and secure.

➡ **UX/UI Designer:**

Role: Designed the user interface and 3D immersive environments.

Responsibility: Conducted user testing to ensure an intuitive and accessible experience.

➡ **Ethics & Compliance Analyst:**

Role: Guided data protection and ethical decision-making.

Responsibility: Ensured compliance with GDPR and university ethics policies.





Existing Skills & Knowledge Development

> Initial Skills (Theoretical):

- Strong academic foundation in our respective fields (CS, Design, Law).
- Basic understanding of project management and ethical frameworks.

> Developed Skills (Practical Application):

- **Technical:** Mastered Unity3D for VR development and Photon for real-time networking.
- **Regulatory:** Applied GDPR principles in a real-world scenario, implementing data anonymization.
- **Project Management:** Gained hands-on experience with Agile methodology and sprint planning.
- **Interpersonal:** Enhanced negotiation and presentation skills through cross-functional work.



Team Strengths, Weaknesses & Solutions

Key Strengths	Key Weakness
➔ Diverse Skill Sets: Complementary expertise covered all project aspects.	➔ Challenge: Initial lack of clarity on handling sensitive biometric data in the VR environment.
➔ Shared Commitment: A unified dedication to creating an ethical, high-quality product.	➔ Action: We proactively sought guidance from the university's Legal and Ethics Department.



Solution: Implemented a robust data anonymization protocol, turning a risk into a core product strength.

Team Communication Practices

✓ Tools & Platforms

- ❑ **Microsoft Teams:** Used for daily virtual stand-up meetings and weekly full-team syncs.
- ❑ **Trello:** Managed our task workflow, providing clear visibility on progress and blockers.
- ✓ **Documentation** - A centralized SharePoint folder ensured all design docs, code, and reports were accessible.
- **Outcome** - Seamless coordination between technical and non-technical members, preventing misunderstandings.



Team Time Management



Methodology: **Agile Framework**

Broke the project into 2-week sprints with specific, achievable goals.

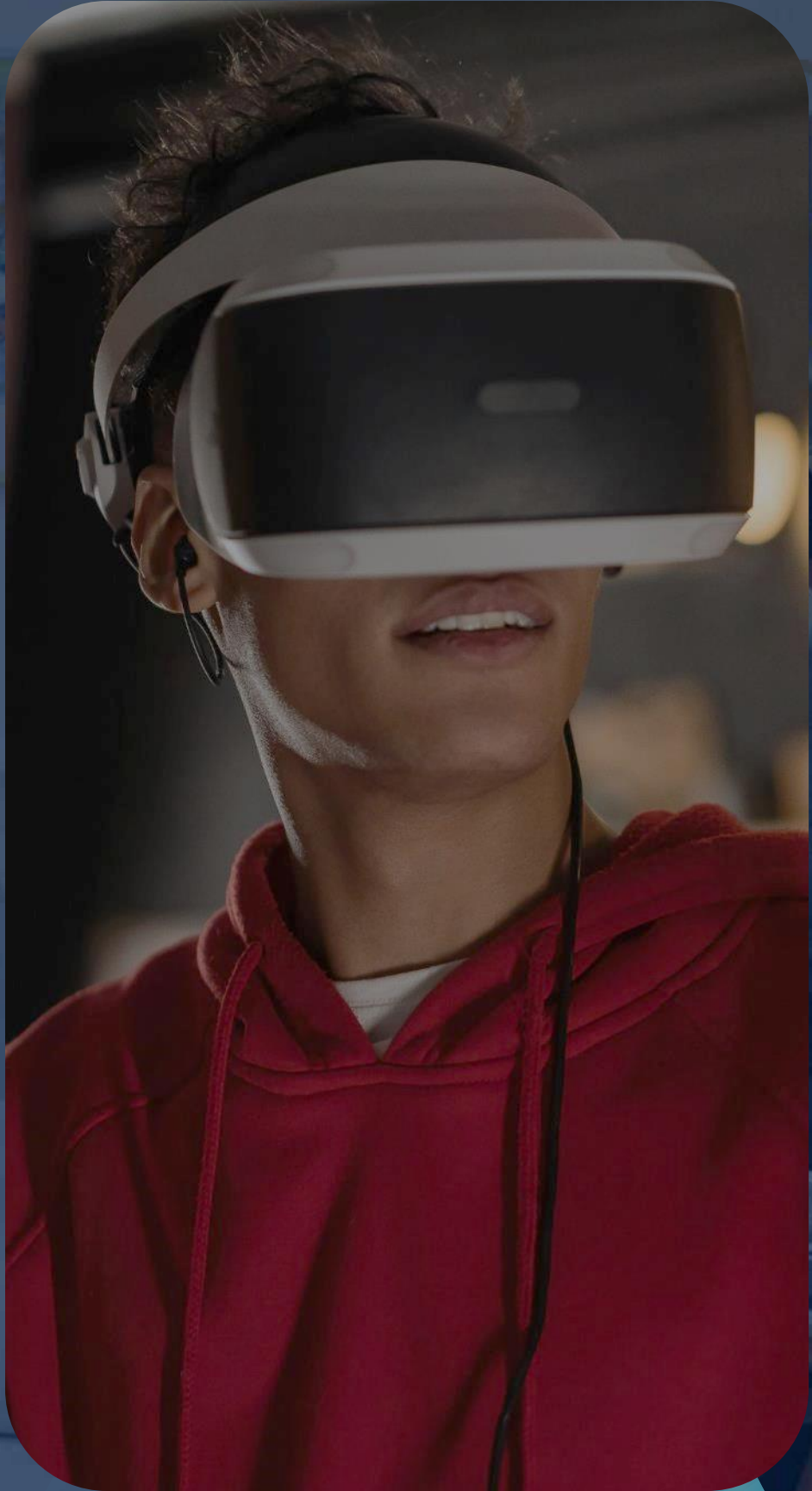
Planning Tools:

- **Gantt Chart:** For high-level milestone tracking against the semester deadline.
- **Trello Board:** For daily task management and prioritization.

Risk Mitigation: Incorporated buffer time into our schedule for unexpected academic pressures.

Result:

All milestones and the final submission were delivered on time.



Actions to Foster Teamwork & Collaboration

Cross-Functional Workshops:

Held joint design-thinking sessions to solve problems creatively.

Blame-Free Retrospectives:

After each sprint, we discussed what went well and what to improve, focusing on processes, not people.

Recognition Culture:

Celebrated individual and team milestones in our Teams channel.

Result:

Built a high-trust environment where members felt comfortable sharing ideas and addressing issues proactively.

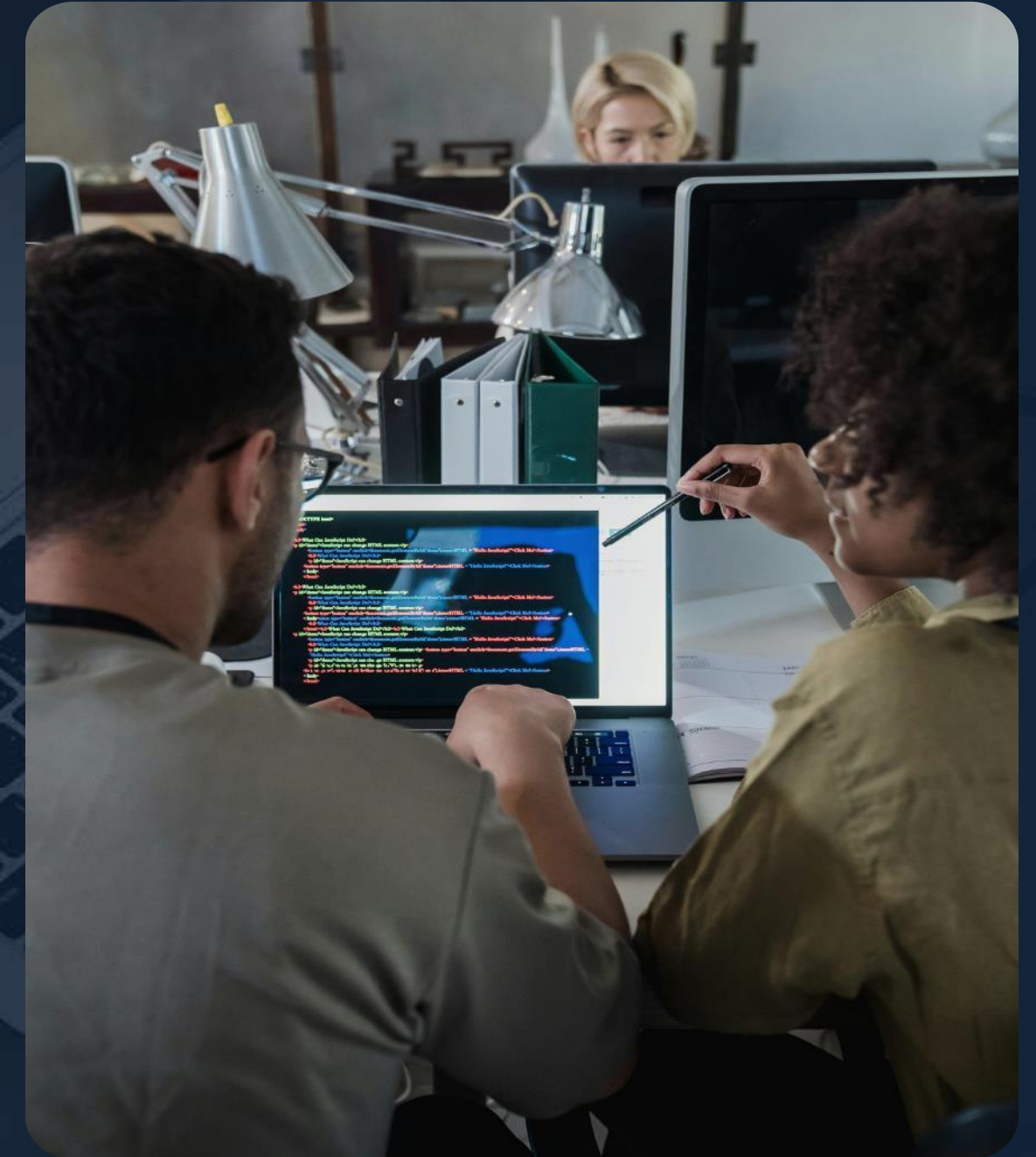
Internal & External Resource Management

Internal Resources (University Support):

- **Legal Department:** Consulted for data privacy compliance.
- **IT Department:** Accessed high-performance computers for VR development and testing.
- **Library Resources:** Utilized academic journals for research on VR in education.

External Resources:

- **Cloud Services:** Selected a GDPR-compliant (EU) cloud provider for hosting.
- **Asset Libraries:** Licensed 3D models and sounds from reputable sources to ensure copyright compliance.



The Implemented Solution: A Snapshot

What We Delivered:

A functional VR classroom prototype for virtual science labs.

Key Features:

- ✓ Immersive 3D environments for chemistry and physics experiments.
- ✓ Real-time avatar-based collaboration for up to 20 students.
- ✓ Instructor dashboard with basic engagement analytics.

Our Pride

The project was not only technically successful but also ethically sound from the ground up.



Our Team's Journey: From Formation to Implementation

1. How Our Team Was (The Beginning) :-

- **Composition:** A newly formed group of individuals with strong, but separate, theoretical knowledge.
- **Dynamic:** Roles were defined, but collaborative processes were untested. We operated as specialists in silos.
- **Initial Challenge:** Integrating our diverse expertise into a single, cohesive workflow.

2. How Our Team Is Now (The Transformation) :-

- **A Cohesive Unit:** Evolved into a high-performing, cross-functional team with proven execution capability.
- **Developed Strengths:** Mastered practical skills in VR development, Agile management, and applied ethics.
- **Key Achievement:** Successfully transformed into a collaborative unit that proactively resolves conflicts and manages time/resources effectively.

3. What Was Implemented(The Outcome) :-

- ✓ **The Deliverable:** A fully functional, ethical VR Classroom prototype for STEM labs.
- ✓ **The Legacy:** A proof-of-concept that demonstrates the successful fusion of technical innovation and strong professional practice.

Key Project Insights

Technical Insight

Building a low-latency, synchronized VR experience is challenging but achievable with the right architecture

Ethical Insight

Proactively addressing privacy concerns adds value and builds trust, it is not a hindrance.

Team Insight

Clear communication and defined processes are the bedrock of successful collaboration, especially in a diverse team.



Recommendations & Future Enhancements

Formalize the Ethical Framework

Propose this "Privacy by Design" approach as a standard for future tech projects within the university.

Continue Skill Development
Recommend advanced VR development workshops for the School of Technologies.

Expand Pilot Program
Seek funding to trial the platform in a live module next academic year.



Conclusion

This project was a comprehensive learning experience in both technical implementation and professional teamwork.

We successfully delivered a proof-of-concept for an ethical VR classroom, demonstrating the potential of emerging technology in education.

Thank you for the opportunity.
We are happy to discuss our journey and the project in more detail.



THANK YOU

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